MILITARY SPECIFICATION

# LATHES, ENGINE AND TOOL ROOM, 16, 20 AND 25 INCH 7-1/2 to 15 HP

This specification is approved for use by all Department and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers electric motor driven, metal turning and threading, engine and toolroom lathes. The lathes shall have either geared head drive or combination gear and belt drive with equipment and accessories as described herein.

1.2 Classification. The lathes covered by this specification shall be of the following types and sizes. The type, size and horsepower to be furnished shall be as specified (see 6.2.1).

Type I - Engine lathes

Type 11 - Toolroom lathes

| Size | Swing over bed Inches | Swing over cross slide Inches |
|---|---|---|
| 1610 | 16 | 10 |
| 2012 | 20 | 12 |
| 2515 | 25 | 15 |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3416
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-government documents which are current on the date of the solicitation.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B5.8 - Chucks and Chuck Jaws.

ANSI B5.9 - Spindle Noses for Tool Room Lathes, Engine Lathes, Turret Lathes, and Automatic Lathes (IS0 702).

ANSI B5.10 - Machine Tapers.

ANSI B5.16 - Accuracy of Engine and Tool Room Lathes.

ANSI B11.6 - Safety Requirements for the Construction, Care, and Use of Lathes.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695. )

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (IS0)

IS0 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)



Source: https://assist.dla.mil -- Downloaded: 202610-19T16:38Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or military standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The lathe shall be new and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The lathe shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Use of a spacer between the headstock and the bed to achieve the capacity required in this specification shall not be acceptable.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (S1) shall be used in the design and construction of the lathe. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6,2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, both the U.S. Customary System of Units (US) and the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the lathe shall be graduated in the same system.

3.2.1.2 Digital readout system. Unless otherwise specified (see 6.2.1), a digital readout system shall be provided on the X and Z axes of travel. The digital readout system shall consist of a position feedback device, an electronic display for each axis and all related components, mounting hard-



Source: https://assist.dla.mil -- Downloaded: 202610-19T16:38Z
Check the source to verify that this is the current version before use.

ware, and electrical equipment necessary to assure complete operation of the system. The digital readout system shall be approved by the Underwriters Laboratory (U.L.). The system shall be capable of providing a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0001 inch. The system accuracy shall be 0.0002 inch per foot and the repeatability shall be 0.0001 inch. Special requirements for digital readout system components shall be as specified (see 6.2.1). The readout system shall be capable of indicating axis location in both the US and SI systems of measurement. The digital readout system shall be provided with a means for calibration.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device or contamination by oil, dust, chips, coolants, etc. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Provisions shall be made to prevent damage to the feedback device in the event of overtravel.

3.2.1.2.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual illuminated digits not less than 0.560 inch high. The console shall be capable of displaying at least six digits simultaneously. The display console shall also provide keyboard entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator’s work position. The console shall be mounted to withstand shock and vibration generated by the lathe. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operation of the lathe. The cable shall be securely fastened to the structure of the lathe in accordance with the requirements of the contract.

3.2.2 Reclaimed materials. The lathe may contain reclaimed materials provided such materials will not jeopardize the machine’s intended use, performance or life. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the lathe.

3.2.3 Energy efficiency. The lathe and its applicable components that directly consume energy in normal operation shall be designed and constructed for the maximum degree of energy efficiency as governed by the lastest developmnets available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.



Source: https://assist.dla.mil -- Downloaded: 202610-19T16:38Z
Check the source to verify that this is the current version before use.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the lathe that present safety hazards. The safety devices shall not interfere with the operation of the lathe. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All lathe parts, components, mechanisms, and assemblies furnished on the lathe, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR 1910 that are applicable to the lathe itself. The machine shall comply with all requirements of ANSI B11.6 that are designated therein as the responsibility of the machine manufacturer. In the event of a conflict between the requirements of OSHA and ANSI standards, the requirements of OSHA shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The lathe shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and-materials containing asbestos shall not be used on or in the machine, or on the provided accessories.

3.2.8 Environmental protection. The lathe shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the lathe shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Enclosed headstock and carriage mechanisms having gears, clutches, sliding parts and bearings that require lubrication shall be serviced by an automatic pressure system, flood system, mist system or any combination of these methods to ensure adequate lubrication to these and all other moving parts. Each system that recirculates lubricating oil shall include a cleanable or replaceable filter. Each lubricant reservoir shall have a convenient means of determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances. When industry or Government standards are applicable, all replacement parts shall conform to the standards. All replacement parts shall be interchangeable with the original respective parts provided by the original equipment manufacturer (OEM) without modification to the replacement part(s) or the mating components . All parts subject to possible future replacement as spare or repair parts shall be assigned a manufacturer’s part number and shall be identified in the repair parts manual. Foreign sourced replacement parts shall be interchangeable with domestically supplied components or be logistically supportable and readily available from a domestic source. Replacement parts are those that are normally subject to wear such as gears, bearings, shafts, feed screws, drive belts, pumps and seals. Electrical replacement parts shall include motors, relays , switches, magnetics, push-



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

buttons, light bulbs, fuses, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts.

3.3 Construction. The lathe shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the lathe to its maximum rating and capacity as specified herein without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3,4 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part .

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the lathe shall be painted in accordance with the manufacturer's commercial practice.

3.3.5.1 Protective finish. When specified (see 6.2.1), the following protective finish requirement shall apply. All surfaces to be painted shall, immediately prior to painting, be cleaned and dried and be free of foreign matter. The protective paint coating shall be at least 0.0020 inches thick and shall have the hardness and adhesion necessary for the prevention of scuffing and peeling. Where no protective finish is utilized (such as bedways, machine guide surfaces, etc.) the natural finish of the material or the finish obtained from heat treatment is permissible provided the surfaces are free from scale or corrosion.

3.3.6 Threads. All threaded parts used on the lathe and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section reference therein.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

3.3.7 Dials. Rotating dials or handwheels shall be used for longitudinal movement or cross motion and shall be graduated in increments not greater than 0.010 inch for longitudinal and 0.001 inch for cross motion. Such graduations shall accurately indicate machine component movement or material removal. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background and be of such size that graduations can be read from the operator's normal position. Feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.3.8 Gears. All gears and pinions of spindle and axis drive trains selected for use in the lathe shall be designed and manufactured in accordance with AGMA 390.03 and AGMA 360.02 for the English (US) or ISO 54 for the metric (SI) system. All back gears, gears in the headstock, gears in the quick change gear box, and gears in the drive train shall be hardened steel. The gears shall be proper width and size to transmit full-rated torque and horsepower through the speed ranges without failure for the expected service life of the machine. The hardness of the gears shall be not less than Rockwell C-48.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. Each lathe shall consist essentially of a bed and base, headstock, headstock spindle, spindle drive, spindle speed change mechanism, carriage and apron, feed mechanism, cross slide, compound rest, and tailstock. In addition, the machine shall have all other equipment specifically described herein and all additional equipment necessary for the machine to meet all of the performance requirements stated herein.

3.4.1 Bed and base. The lathe shall have a bed and base structure that supports the headstock, tailstock, carriage and other related equipment. The bed shall be a stress relieved ferrous casting with sufficient strength and rigidity to maintain accurate alignment of all the components in accordance with ANSI B5.16. The bed shall have V-ways or flat ways or a combination of "V" and flat ways to provide a sliding surface for t-he longitudinal movement of the carriage and assure alignment of the headstock and tailstock. The bed ways shall be either cast integrally with the bed structure or replaceable way inserts, or a combination thereof. The bed ways for the carriage longitudinal movement shall be hardened to Rockwell C-5C if integral or Rockwell C-55 if inserted, and ground to minimize wear and maintain accuracy of alignment. Means shall be provided to lubricate the bed ways automatically or with a one-shot lubrication method as specified (see 6.2.1). The bed shall have a sufficient number of covered access openings to facilitate inspection, adjustment, and repair of components located within the structure. The base shall be provided with means for securing and leveling the machine on a foundation.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

3.4.2 Chip and coolant pan. The chip and coolant pan shall allow coolant to drain from the chips and be returned to the coolant system for reuse. There shall be adequate clearance between the chip pan and the lathe bed to allow for chip removal.

3.4.3 Headstock. The lathe shall have a headstock that supports and aligns the spindle axis parallel with the bed ways. It shall contain all the gearing necessary to provide the required speeds; clutch and quick acting brake mechanism to start and stop the spindle rotation and to allow free rotation for changing speeds; and a power takeoff to the gearbox for feeding and threading. The headstock shall house the gearing for the spindle drive and the power take-off source for the feed mechanism. Speed selections shall be by manually setting levers or dials on the headstock. Spindle start and stop shall be by levers on the headstock and on the apron of the lathe. The headstock shall be constructed to support the spindle and gear shafts so as to prevent deflection that would affect accuracy under full operating and cutting loads. All load carrying surfaces of gears, clutch jaws and surfaces of shafts on which' gears and clutches slide, or internal surfaces of sliding parts, shall have a surface hardness of not less than Rockwell C-48. When power to the spindle drive mechanism is transmitted through V-belts, means shall be provided to maintain proper belt tension. The headstock shall be oiltight and shall have automatic lubrication to all internal parts that require lubrication. When specified (see 6.2.1), a temperature sensor shall be provided to monitor the temperature of the headstock lubricant and shut the machine off at a preset temperature limit below the flashpoint of the oil vapor.

3.4.4 Headstock spindle. The lathe shall have a headstock spindle to provide accurate mounting and rotation of a chuck, faceplate, or collet. The spindle shall be supported by roller or ball type bearings of a size, type and precision class suitable for the application. Unless otherwise specified (see 6.2.1), the spindle nose type shall be type "D", as normally provided and as specified in table I, and shall conform to the requirements of ANSI B5.9. The spindle internal taper shall be as specified in table I and shall conform to the requirements of ANSI B5.10. The spindle nose shall be accurately machined to provide locating surfaces for lathe centers, faceplates and chucks. The inner and outer surfaces of the spindle nose on which work holding devices are mounted shall be hardened to at least Rockwell C-40.

3.4.5 Spindle drive. The lathe shall have a spindle drive mechanism that will provide the required range of spindle speeds listed in table I. The spindle shall be driven through an all gear or combination gear and belt drive by an electric motor as specified in table I. All gears shall meet the requirements of 3.3.8.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

3.4.6 Spindle speed change mechanism. The spindle speed mechanism shall be the manual speed change type, the manual speed change with pre-select power shift type or the variable speed type as specified (see 6.2.1.) The shifting mechanism shall be designed so that speed change cannot occur unintentionally. Except for the variable speed drive, speeds shall be in geometric progression within 10 percent. The spindle shall operate in both directions of rotation. The manual speed change mechanism shall allow spindle speed changes by positioning gear shift levers or dials located on the front side of the headstock. Speed shall be selected by positioning the levers as indicated on a conveniently located speed chart or by positioning a direct reading dial. The manual speed change with preselect power shift shall provide spindle speed selections by manually setting a selector control to the desired spindle speed. It shall be possible to preselect the next spindle speed while still cutting, then automatically initiate the shift upon activation of the spindle control. The variable speed change mechanism shall permit spindle speed adjustments to compensate for varying machine conditions. Optimum speeds shall be obtained by gear range selection and by variable speed adjustment within the range. It shall be possible to adjust speed within a gear range while running. The actual spindle speed in RPM shall be shown by a speed indicating device located convenient to the operator.

3.4.7 Feed thread gearbox. Unless otherwise specified (see 6.2.1), the lathe shall be equipped with a selective type, combination inch and metric gearbox to provide a full range of inch and metric feeds and threads as specified in table I, without the use of transposing gears. The cross slide feed screw and the compound slide feed screw shall each be equipped with dials graduated to read measurements in both the inch system and the metric system. Power feeds and thread leads shall be provided by the totally enclosed quick change gearbox. Longitudinal and cross feeds shall be provided by a feed rod from the gearbox to a feed mechanism within the apron. Feeds shall be in either direction regardless of the direction of spindle rotation. Threading shall be done using a leadscrew from the gearbox and a split nut mechanism within the carriage apron. The lathe shall be able to cut either right-hand or left-hand threads regardless of the direction of spindle rotation. The gearbox shall be automatically lubricated. Safety clutches or other devices shall protect the feed/thread gearbox, feed rod and leadscrew from accidental overload. An interlock shall prevent simultaneous engagement of the feed and thread mechanism. A feed and thread chart shall be mounted near the gear box as a ready reference for feed and thread selections. All feed and thread selections shall be shown without need for mathematical conversion.

3.4.8 Carriage. The lathe shall have a carriage that supports the apron, cross slide, compound rest, and other related equipment. The carriage shall traverse longitudinally on the bed ways. The carriage shall be fitted to the bed ways and shall have means of compensating for wear between sliding surfaces. Means shall be provided to lock the carriage to the bed at any desired location between the headstock and tailstock for cross feed type operations . The carriage shall operate in both longitudinal directions



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

under power, regardless of the direction of spindle rotation. The carriage wings shall have means to prevent chips and abrasive materials from entering between the carriage slide and bed ways. Provision shall be made for use of a taper attachment and follow rest . The top of the carriage shall have ways perpendicular to the axis of the machine to support the cross slide and compound rest.

3.4.9 Apron. The lathe shall have an apron on the front side of the carriage which all contain the feed mechanism, threading mechanism, operating controls and lubrication system for the apron and carriage components. The operating controls shall include power longitudinal and cross feeds, automatically operated longitudinal feed stops, threading controls, and a spindle start and stop. Lever type operating controls such as the threading and feed engagement shall fully engage and disengage with smooth, positive action. The apron shall be double wall construction providing inboard and outboard supports for both ends of gear shafts. Gearing in the apron shall provide hand and power longitudinal and cross feeds. An interlock shall be incorporated in the apron to prevent simultaneous engagement of the feed and half-nut. Handwheels shall automatically disengage to assure nonrotation while the carriage is under power. The feed rod and leadscrew shall be separate units.

3.4.9.1 Leadscrew reverse mechanism. When specified (see 6.2.1), the lathe shall have a leadscrew reversing mechanism to permit reversing the direction of the carriage without disengaging the leadscrew half-nut or changing the speed and direction of spindle rotation. The control for leadscrew reversing shall be conveniently located to the operators normal work station.

3.4.10 Cross slide. The cross slide shall be fitted to the carriage ways and shall have means to compensate for wear between the sliding surfaces. The cross slide shall be positioned by a precision feed screw which shall be protected from dirt and chips.

3.4.11 Compound rest. A compound rest shall be mounted on top of the cross slide. The base of the compound rest shall be graduated in one degree increments to not less than 90 degrees each side of a zero reference. The base shall swivel horizontally through 360 degrees. Means shall be provided to lock the compound at any angular setting. The compound rest shall have a T-slot for mounting tool holders. The compound slide feed screw shall be protected from dirt and chips.

3.4.12 Tailstock. The tailstock shall be constructed to permit supporting a work part close to the compound rest without interference with operation of the top slide. The tailstock shall be moved easily along the bedways and have a quick acting clamp for locking in position. The tailstock spindle shall hold and eject tang type tools, and shall be graduated in increments not greater than 1/16 inch for measuring spindle extension. The tailstock spindle taper shall be in accordance with ANSI B5.10 for self-holding tapers. The tailstock shall be marked to indicated set-over as specified in table I. Tailstock spindle locking shall not affect accuracy of alignment.

3.4,13 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

shall have means for draining and cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.14. Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each lathe shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The lathe shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating. The wiring in each machine shall have individual point-to-point color coded wires in conformance with ANSI/NFPA 79. Diagrams/schematics shall clearly identify the individual, point-to-point color coded wire and wire termination point location in the electrical system.

3.4. 14.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-hertz (Hz) unless otherwise specified (see 6.2.1). Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a drip-proof enclosure, and the requirements of ANSI/NEMA MG1. Motor horsepower ratings shall conform to or exceed the requirements of table I.

3.5 Sizes and capacities. Unless otherwise specified (see 6.2.1), the size and characteristics of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Alignment and performance accuracies . Each lathe shall meet the alignment and performance accuracies set forth in table II using the tests described in ANSI B5.16, “Accuracy of Engine and Tool Room Lathes.” Type I lathes shall meet or exceed the values found under the heading “Engine Lathes” under the proper size for the lathe specified. Type II lathes shall meet or exceed the values found under the heading “Tool Room Lathes.”

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), each machine shall be furnished with all standard equipment normally furnished by the machine manufacturer. The following shall also be included as standard equipment.

3.7.1 Lathe centers and center sleeve. One headstock center and one tailstock–center shall be furnished. One headstock center sleeve shall be provided for adapting the taper in the spindle nose to the taper of the center supplied. The centers shall be hardened to Rockwell C-62 to C-66. The centers and the sleeve shall be ground to within 0.0002 inch total indicator reading for concentricity when inserted in the headstock and checked for concentricity over their entire lengths.

3.7.2 Chuck and chuck jaws. The number and type of chuck and chuck jaws shall be as specified (see 6.2.1), shall be steel and shall conform to the applicable requirements of ANSI B5.8. A self-ejecting chuck key shall be provided for each chuck.



Source: https://assist.dla.mil -- Downloaded: 202610-19T16:38Z
Check the source to verify that this is the current version before use.

3.7.3 Drive plate. The drive plate shall be the maximum allowable diameter, not to exceed the nominal swing over the bed. The drive plate shall be for mounting on the spindle nose and shall have at least one radial slot for driving a lathe dog.

3.7.4 Taper attachment, The taper attachment shall mount on the rear of the carriage. It shall be graduated for settings in both degrees and inches of taper per foot. Means for maintaining adjustment shall be provided. The attachment shall turn either right or left hand tapers. The taper attachment shall not interfere with the normal operation of the lathe.

3.7.5 Thread-chasing dial. An inch/metric thread-chasing dial shall be furnished which will allow disengaging of the half-nuts and re-engaging during threading for both inch and metric lead threads. The thread chasing dial shall be mounted on the carriage apron and be driven by the leadscrew.

3.7.6 Micrometer carriage stop. The micrometer carriage stop shall be mounted on the front bed to provide an adjustable stop. The micrometer dial shall be graduated in increments of 0.001 inch or less with at least 0.750 inch of adjustment.

3.7.7 Work light. Unless otherwise specified (see 6.2.1), a work light shall be provided. The work light shall be a complete unit with all necessary equipment for mounting on the carriage and shall be adjustable to provide maximum lighting of the work area. The lamp shall be shielded against dirt and chips. The lamp shall operate on not more than 130 volts.

3.7.8 Wrenches. All special tools and wrenches required for operating the lathe and accessories that are normally supplied with the lathe shall be provided.

3.8 Optional equipment. Optional equipment shall be furnished as specified and as fully described (see 6.2.1).

3.9 Additional requirements. Performance features, attachments or accessories required in addition to, or in lieu of those specified herein, shall be as required by the procuring activity (see 6.2.1).

3.10 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a corrosion-resistant lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval for each point
Type of lubricant required for each point



Source: https://assist.dla.mil -- Downloaded: 202610-19T16:38Z
Check the source to verify that this is the current version before use.

Viscosity of each lubricant
Military or Federal specification number of each lubricant
Size and type of lubricant filter

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard lathe and a suffix identified in the manufacturer's permanent records, The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Type Size Horsepower
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. Unless otherwise specified (see 6.2.1), the lathe shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the lathe. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.12 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the lathe and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such



Source: https://assist.dla.mil -- Downloaded: 202610-19T16:38Z
Check the source to verify that this is the current version before use.

inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5, The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it comment the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under normal operating conditions.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2,1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article. examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6,2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1 through 4.7.2.3, and the inspection in 4.8. When a protective finish (paragraph 3.3.5.1) is required, (see 6.2.1 j) the test described in 4.7.4 shall be performed. When electromagnetic interference control (paragraph 4.7.3) is required (see 6.2.1.k) the test described in 4.7.3 shall be performed. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.6 Examination. The lathe shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. Each lathe shall be operated at no load for not less than 60 minutes. Proper operation of all controls, motors, adjusting mechanisms , and accessories shall be verified during the trial period.



Source: https://assist.dla.mil -- Downloaded: 202610-19T16:38Z
Check the source to verify that this is the current version before use.

4.7.1.1 Alignment tolerance test. Each lathe shall be tested to determine that it complies with the alignment tolerance requirements of paragraph 3.6. Following completion of the tests described in paragraphs 4.7.1 through 4.7.2.3 the alignment tolerance test shall be repeated to determine that the lathe remains in compliance with the requirements of paragraph 3.9 without requiring adjustments.

4.7.2 Performance test. Each lathe shall be subjected to the following cutting tests to determine whether the performance under cutting loads is within the requirements of this specification. When specified (see 6.2.1), tests of specific requirements shall be performed in addition to or in lieu of the following tests.

4.7.2.1 Round turning test. A test bar of SAE 1040 steel, or steel of equal machinability, shall be held in a chuck and turned for a length of approximately two inches at the RPM and feed rate based on the recommended surface cutting speed for the material and tool used. The finish cut shall be conducted in accordance with test number 21 of ANSI B5.16 and shall meet the tolerances as stated in table II of this specification under the proper category for the type lathe specified.

4.7.2.2 Cylindrical turning test. A test bar of SAE 1040 steel, or steel of equal machinability, shall be rough turned to the configuration of the specimen in ANSI B5.16 for cylindrical turning. A finish cut using carbide tooling shall be made over the two collars in one pass without adjustment between collars. The turned collars shall be within the accuracy tolerance specified in test number 22 of table II under the proper category for the type lathe ordered. The test part shall then be prepared for turning between centers as described in test number 23 of ANSI B5.16. A cut shall be made over the two collars and the diameters shall be within the tolerance for turning cylindrical between centers in test number 23 of table II. Spindle speeds and feeds shall be based on the recommended surface cutting speed for the material and carbide tooling.

4.7.2.3 Roughing test, A square specimen of 1040 steel or steel of equal machinability with 3-inch sides, and 16 inches in length shall be rough-turned between centers by two or more cuts; depth of cut 0.250 inch, and a cutting speed of approximately 40 rpm, and feed rate of approximately 0.035 inches . The maximum diameter of the bar at the tailstock and the maximum diameter of the bar measured 10 and 15 inches from the tailstock end of the bar, shall not vary by more than plus or minus 0.001 inch, The kickback of the carriage relative to the bed shall be measured and shall not exceed 0.001 inch during any single cut. The deflection of the tailstock spindle shall not exceed 0.002 inch, measured l/8-inch from the nose of the spindle during the roughing test.



Source: https://assist.dla.mil -- Downloaded: 202610-19T16:38Z
Check the source to verify that this is the current version before use.

4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control (see 6.2.1.k) shall be tested for compliance with 3.3.9.

4.7.4 Protective finish test. When specified (see 6.2.1), equipment requiring a protective finish shall be tested for compliance with 3.3.5.1. The hardness, adhesion and thickness of the protective finish on the lathe shall be checked as follows:

a. The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the lathe until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.

b. A test sample of the same metal, painted with the same process and at the same time as the lathe, shall have a sharpened H lead pencil pressed into the paint as in (a). Failure to penetrate the surface of the paint down to the base metal, (paint Loo hard), shall be cause for rejection.

c. Two, 2-inch intersecting lines shall be cut through the paint to the base metal of the test sample. A strip of 1/2-inch wide fiberglass tape shall be firmly pressed over the Intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.

d. Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least 0.0020 inch-thick shall be cause for rejection.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

## 5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

## 6. NOTES

6.1 Intended use. The lathes covered by this specification are intended for use—in any machine shop for production or toolroom where the nature of the work performed does not require heavy machining capabilities. The lathes are suitable for turning, boring, facing, threading, and taper turning using high-speed steel or carbide tooling.



Source: https://assist.dla.mil -- Downloaded: 202610-19T16:38Z
Check the source to verify that this is the current version before use.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, size and horsepower required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Digital readout system, if not required (see 3.2.1.2).

g. Special requirements for digital readout, if required (see 3.2.1.2).

h. Additional safety and health requirements, if required (see 3.2.5).

i. Painting, if different (see 3.3.5).

j. Protective finish, if required (see 3.3.5.1).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

l. Type of bedway lubrication, specify automatic or one shot (see 3,4.1).

m. Headstock temperature sensor, if required (see 3.4.3).

n. Spindle nose, if different (see 3.4.4).

0. Spindle speed mechanism type, as specified (see 3.4.6).

p. If either English or metric gearbox is required (see 3.4.7).

q. Leadscrew reverse mechanism, if required (see 3.4.9.1).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

r. Electrical system, if different (see 3.4.14).

s. Specify if AC motors to be designed for other than 60 Hz (see 3.4.14.1).

t. Type motor enclosure, if different (see 3.4.14.1).

u. Size and capacity, if different (see 3.5).

v. Standard equipment, if different (see 3.7).

w. Specify chuck and chuck jaws required (see 3.7.2).

x. Work light, if different (see 3.7.7),

y. Optional equipment, as required (see 3.8).

z. Additional requirements, as required (see 3.9).

aa. Lubrication chart or plate, if different (see 3.10.1).

bb. Nameplate, if different (see 3.10.2).

cc. Hour meter, if not required (see 3.11).

dd. First article inspection, if different (see 4.4).

ee. Quality conformance inspection, if different (see 4.5).

ff. Performance tests, if different fully describe (see 4.7.2).

gg. Protective finish test, if required (see 4.7.4).

hh. Specific packaging, packing and special marking, if required (see 5.1). (Specify level A packaging, specify level A or B packaging and specify special marking if required. )

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

TABLE I - Size and capacities.

| Characteristics | Type I and Type II Size - Dimensions in inches | | |
|---|---|---|---|
| | 1610 | 2012 | 2515 |
| Swing over bed, inches | 16 | 20 | 25 |
| Swing over cross slide, inches | 10 | 12 | 15 |
| Distance between centers, inches: | | | |
| Standard | 40 | 40 | 72 |
| Optional | 54, 78, 100, 120, 144, 200 | 54, 60, 80, 100, 120, 160, 200 | 96, 144, 240 |
| Spindle speed selections: | | | |
| Standard | 12 | 12 | 12 |
| Optional | 16 | 16 | — |
| Spindle speed ranges, RPM: | | | |
| Standard | 40 to 1400 | 40 to 1400 | 40 to 1400 |
| Optional | 32 to 1000 | 32 to 1600 | — |
| Cross slide travel, inches (w/o taper) | 10 | 10 | 13 |
| Compound slide travel, inches | 4 | 4 | 4-3/4 |
| Main drive motor, H.P: (continuous) | | | |
| Standard | 7-1/2 | 12 | 12 |
| Optional | 10 | 12-1/2 or 15 | 15 |
| Size hole through spindle, inches | 2-1/8 | 2-1/2 | 3 |
| Spindle nose size, type | 6", 8", D-1 | 6", 8", 11", D-1 | 8", 11" D-1 |
| Spindle center, Morse taper | #4 | #4 | #5 |
| Longitudinal feed selections | 45 | 28 | 45 |
| Longitudinal feeds, inches per revolution (IPR) | .0025 to .028 | .0025 to .028 | .0036 to .104 |
| Threads, English: | | | |
| Number of selections | 38 | 38 | 38 |
| Number of threads, threads per inch (TPI) | 3 - 42 | 3 - 42 | 3 - 42 |
| Threads, Metric: | | | |
| Number of selections | 32 | 30 | 40 |
| Number of threads, mm | .4 to 14 | .4 to 14 | .4 to 14 |
| Tailstock spindle diameter, inches | 2-3/8 | 2-7/16 | 3-1/8 |
| Tailstock spindle travel, inches | 6 | 6 | 7-1/2 |
| Tailstock center, Morse taper | #4 | #5 | #5 |
| Tailstock set over, each side of center, inches | 1/2 | 3/16 | 1/2 |
| Leadscrew diameter, inches | 1-3/16 | 1-3/16 | 1-7/16 |

NOTE: Machine size and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

TABLE II. Alignment tolerances.

| ANSI B5.16 test number | Test (Dimensions in inches) | Toolroom lathe | Engine lathe 2012 | Engine lathe 2515 |
|---|---|---|---|---|
| 1 | Bed level, transverse direction, in 12 inches | +0.00024 | +0.00024 | +0.00024 |
| 2 | Bed level, longitudinal direction, in 12 inches | within 0.00024 | within 0.00024 | within 0.00024 |
| 3 | Tailstock way alignment, in 12 inches | within 0.00024 | within 0.00024 | within 0.00024 |
| 4 | Spindle center runout | 0 to 0.0004 TIR | 0 to 0.0004 TIR | 0 to 0.0004 TIR |
| 5 | Spindle nose runout | 0 to 0.0003 TIR | 0 to 0.0004 TIR | 0 to 0.0004 TIR |
| 6 | Cam action of spindle | 0 to 0.0003 TIR | 0 to 0.0005 TIR | 0 to 0.00075 TIR |
| 7 | Spindle taper runout, at end of 12-inch test bar | 0 to 0.0006 TIR | 0 to 0.00012 TIR | 0 to 0.0012 TIR |
| | at end of spindle nose | 0 to 0.0003 TIR | 0 to 0.0004 TIR | 0 to 0.0004 TIR |
| 8 | Headstock alignment vertical, in 12 inches | 0 to 0.0005 | 0 to 0.0008 | 0 to 0.0008 |
| 9 | Headstock alignment horizontal in 12 inches | 0 to 0.0002 | 0 to 0.0008 | 0 to 0.0008 |
| 10 | Tailstock spindle alignment horizontal in 4 inches | 0 to 0.0004 | 0 to 0.0004 | 0 to 0.0004 |
| 11 | Tailstock spindle alignment vertical, in 4 inches | 0 to 0.0008 | 0 to 0.0008 | 0 to 0.0008 |
| 12 | Tailstock taper alignment horizontal, in 12 inches | 0 to 0.0004 | 0 to 0.0008 | 0 to 0.0008 |
| 13 | Tailstock taper alignment vertical, in 12 inches | 0 to 0.0004 | 0 to 0.0012 | 0 to 0.0012 |
| 14 | Vertical alingment of head and tail centers | 0 to 0.0008 | 0 to 0.0008 | 0 to 0.0008 |
| 15 | Readings for base length bed taken with lead screw stationary — Vertical | 0 to 0.0004 | 0 to 0.0004 | 0 to 0.0004 |
| | Horizontal | 0 to 0.0004 | 0 to 0.0004 | 0 to 0.0004 |
| | Alignment of half-nut horizontal or vertical | 0 to 0.006 | 0 to 0.006 | 0 to 0.006 |
| 16 | Lead screw cam action | 0 to 0.0002 | 0 to 0.0004 | 0 to 0.0004 |
| 17 | Cross-slide alignment, on 12-inch diameter | 0 to 0.0005 | 0 to 0.0008 | 0 to 0.008 |
| 18 | Face plate runout — On diameter | 0 to 0.0005 | 0 to 0.001 | 0 to 0.001 |
| | On face at diameter | 0 to 0.001 | 0 to 0.0015 | 0 to 0.00015 |
| 19 | Chuck runout — Face and periphery | 0 to 0.003 | 0 to 0.003 | 0 to 0.003 |
| | Face of steps | 0 to 0.003 | 0 to 0.003 | 0 to 0.003 |
| | Bar test 3 inches from end of jaw | 0.003 maximum | 0.003 maximum | 0.003 maximum |
| 20 | Collet chuck runout | 0 to 0.001 | 0 to 0.001 | 0 to 0.001 |
| 21 | Lathe must turn round with work mounted in chuck | 0 to 0.0002 | 0 to 0.0004 | 0 to 0.0004 |
| 22 | Lathe must turn cylindrical, 12-inch workpiece mounted in a chuck | 0 to 0.0008 | 0 to 0.0012 | 0 to 0.0012 |
| 23 | Lathe must turn cylindrical, 12-inch workpiece mounted on centers | 0.0004 | 0.0008 | 0.0008 |
| 24 | Leadscrew, lead per 12 inches | +0.0001 | +0.0015 | +0.002 |
| | Lead in any 4 inches | ∓0.0004 | ∓0.0005 | ∓0.0007 |
| 25 | Backlash on cross-feeds screw | ¯0.004 | ¯0.004 | ¯0.005 |
| | on compound rest screw | 0.004 | 0.004 | 0.005 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

6.4 Subject term (key word) listing.

Lathe, Engine
Lathe, Tool Room
Turning

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Army - ME, SM
Navy - AS, MC, YD

Preparing activity:
DLA - IP

project (3416-0203)



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 4966 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

DEFENSE LOGISTICS AGENCY
DEFENSE INDUSTRIAL PLANT EQUIPMENT CENTER
ATTN: DIPEC-SSM
MEMPHIS, TN 38114-5297

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-L-23400E | Lathes, Engine and Tool Room, 16, 20 and 25 Inch 7-1/2 to 15 HP |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

*(TO DETACH THIS FORM, CUT ALONG THIS LINE.)*

Source: https://assist.dla.mil -- Downloaded: 202640-19T16:38Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:38Z
Check the source to verify that this is the current version before use.